SE444 Mock Final Exam Grading Guide – Fall 2025

Total: 30 points (Part I: 6pts, Part II: 11pts, Part III: 5pts, Part IV: 6pts, Part V: 5pts)

Part I: Multiple Choice (6 pts)

Partial credit: 1pt all correct, 0.5pt if 1 error, 0pt if 2+ errors

Q1 (d-Separation, 1pt): A, D

- A: $A \perp B$ unconditionally (collider C blocks path)
- D: $A \perp D|C$ (chain blocked when middle observed)
- *NOT B,C*: Observing C or D opens collider path
- *NOT E*: $C \rightarrow D$ are directly dependent

Q2 (Bayes, 1pt): B, C, D

- $P(I|A) = 0.0095/0.059 \approx 0.161 < 0.5$
- B: TRUE – posterior ¡ likelihood
- C: TRUE – base rate (1%) dominates
- D: TRUE – with $P(I) = 0.10$, posterior $\approx 0.68 > 0.5$

Q3 (MDPs, 1pt): A, B, C, D

- A: $\gamma = 0$ means myopic (immediate rewards only)
- B: Bellman converges for $\gamma < 1$
- C: VI and PI find same optimal policy
- D: Higher γ = more farsighted
- *NOT E*: γ can change optimal policy

Q4 (Game Theory, 1pt): A, C, D, E

- A: Both have dominant strategies (Row: Down, Col: Right)
- C: (Down,Right) is NE – neither can improve
- D: Down dominates Up for Row (4¿3 vs Left, 2¿1 vs Right)
- E: NE (2,2) ¡ (Up,Left) (3,3) for both players

Q5 (Planning, 1pt): A, B, D, E

- A: Forward search = progression from initial
- B: CWA = unstated atoms false
- *NOT C*: Delete list removes At(C1,Depot)
- D: Backward can be more efficient
- E: STRIPS = deterministic, fully observable

Q6 (Code, 1pt): A, C, E

- A: UCS (priority = cost only, no heuristic)
- C: Goal test on expansion (pop)
- E: Counter breaks priority ties
- *NOT B*: No heuristic in code
- *NOT D*: UCS $\neq$ BFS implementation

Part II: Probability & BN (11 pts)

Q1 (BN, 4pts):

- (a) 2pts: Correct arrows + identify colliders N, L
- (b) 1pt: NO independence (observing F opens path)
- (c) 1pt: Collider identification + explaining away example

Q2 (Inference, 4pts):

- (a) 2pts: Correct $P(H|L) = 0.902$ via enumeration
- (b) 2pts: Correct VE with factor tables shown
- *Partial*: 1pt each for correct setup but calculation error

Q3 (MEU, 3pts):

- (a) 2pts: EU(Fly)=73, EU(Detour)=66, EU(Wait)=53
- (b) 1pt: Fly Direct with justification

Part III: MDPs (5 pts)

Q1 (MDP+VI, 5pts):

- (a) 1pt: All 5 components correct
- (b) 2pts: 1pt per state for correct VI iterations
- (c) 2pts: 1pt policy + 1pt justification
- *Accept*: Minor calculation differences due to rounding

Part IV: Planning & GT (6 pts)

Q1 (STRIPS, 3pts):

- (a) 1pt: Load action with Pre/Add/Del
- (b) 2pts: Valid plan sequence (accept variations)

Q2 (Nash, 3pts):

- (a) 1pt: Premium dominated by Budget
- (b) 2pts: Both NE identified: (Budget,Budget) and (Freemium,Freemium)

Part V: Project Code (5 pts)

Q1 (A*, 3pts): 1pt each for (a), (b), (c)

Q2 (Bayes, 2pts):

- (a) 1pt: Posterior ≈ 0.794
- (b) 1pt: Posterior = Prior (uninformative)

Common Mistakes to Watch:

- **Q1 (Part I):** Confusing collider behavior when observed vs. unobserved
- **Q2 (Part I):** Base rate neglect – assuming high likelihood means high posterior
- **Q2 (Part II):** Not showing factor tables in Variable Elimination
- **Q1 (Part III):** Forgetting living reward (-1) in Bellman equation
- **Q2 (Part IV):** Only finding one Nash Equilibrium (there are TWO)
- **Q2 (Part V):** Not explaining why accuracy=0.5 gives no information